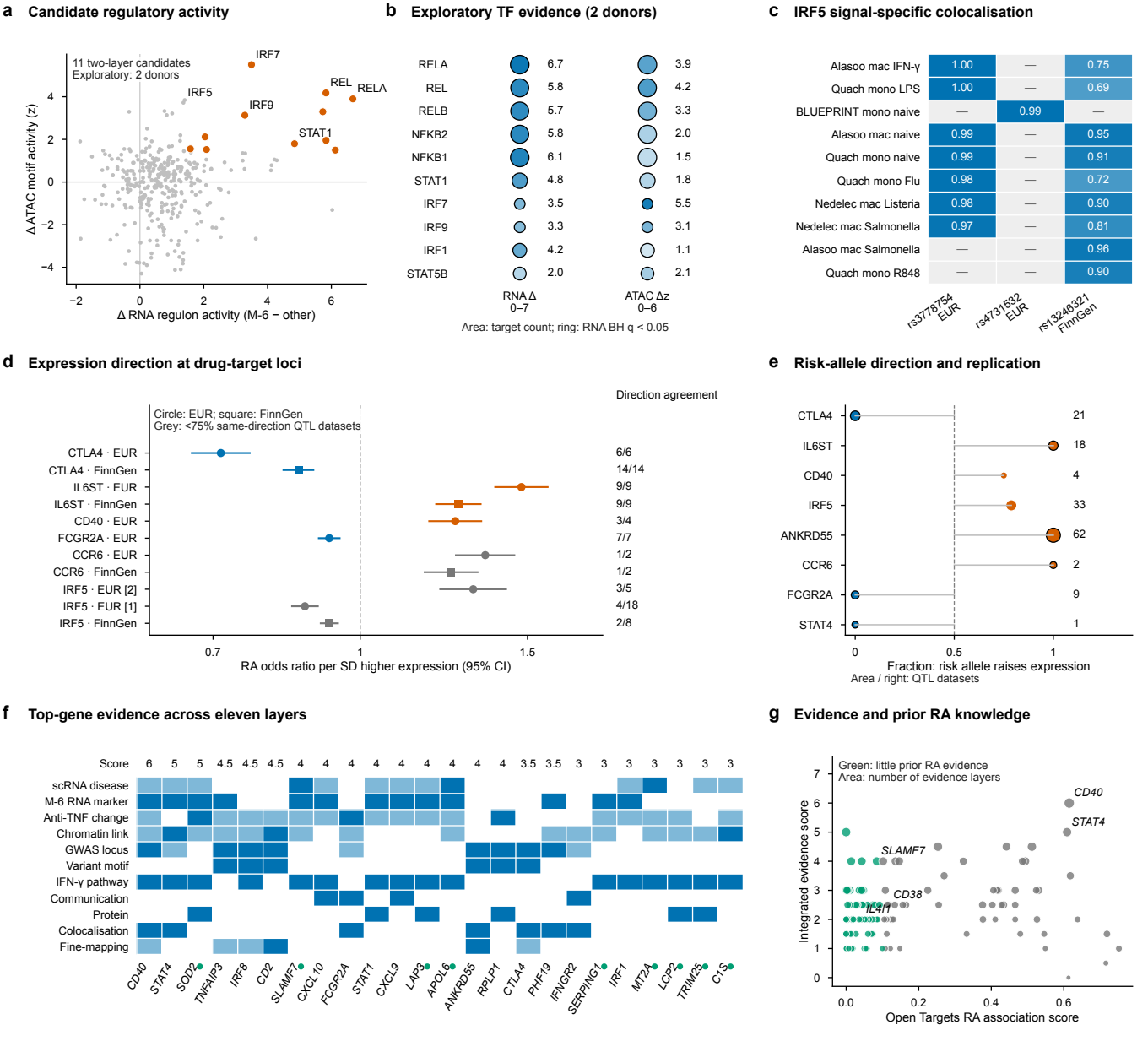

a
Candidate regulatory activity
11 two-layer candidates
Exploratory: 2 donors
IRF7
REL
RELA
IRF5
IRF9
STAT1
Δ ATAC motif activity (z)
Δ RNA regulon activity (M-6 − other)
b
Exploratory TF evidence (2 donors)
RELA 6.7 3.9
REL 5.8 4.2
RELB 5.7 3.3
NFKB2 5.8 2.0
NFKB1 6.1 1.5
STAT1 4.8 1.8
IRF7 3.5 5.5
IRF9 3.3 3.1
IRF1 4.2 1.1
STAT5B 2.0 2.1
RNA Δ 0–7
ATAC Δz 0–6
Area: target count; ring: RNA BH q < 0.05
c
IRF5 signal-specific colocalisation
Alasoo mac IFN-γ 1.00 — 0.75
Quach mono LPS 1.00 — 0.69
BLUEPRINT mono naive — 0.99 —
Alasoo mac naive 0.99 — 0.95
Quach mono naive 0.99 — 0.91
Quach mono Flu 0.98 — 0.72
Nedelec mac Listeria 0.98 — 0.90
Nedelec mac Salmonella 0.97 — 0.81
Alasoo mac Salmonella — — 0.96
Quach mono R848 — — 0.90
rs3778754 EUR
rs4731532 EUR
rs13246321 FinnGen
d
Expression direction at drug-target loci
Circle: EUR; square: FinnGen
Grey: <75% same-direction QTL datasets
Direction agreement
CTLA4 · EUR 6/6
CTLA4 · FinnGen 14/14
IL6ST · EUR 9/9
IL6ST · FinnGen 9/9
CD40 · EUR 3/4
FCGR2A · EUR 7/7
CCR6 · EUR 1/2
CCR6 · FinnGen 1/2
IRF5 · EUR [2] 3/5
IRF5 · EUR [1] 4/18
IRF5 · FinnGen 2/8
RA odds ratio per SD higher expression (95% CI)
e
Risk-allele direction and replication
CTLA4 21
IL6ST 18
CD40 4
IRF5 33
ANKRD55 62
CCR6 2
FCGR2A 9
STAT4 1
Fraction: risk allele raises expression
Area / right: QTL datasets
f
Top-gene evidence across eleven layers
Score 6 5 5 4.5 4.5 4.5 4 4 4 4 4 4 4 4 4 3.5 3.5 3 3 3 3 3 3 3 3
scRNA disease
M-6 RNA marker
Anti-TNF change
Chromatin link
GWAS locus
Variant motif
IFN-γ pathway
Communication
Protein
Colocalisation
Fine-mapping
CD40 STAT4 SOD2 TNFAIP3 IRF8 CD2 SLAMF7 CXCL10 FCGR2A STAT1 CXCL9 LAP3 APOL6 ANKRD55 RPLP1 CTLA4 PHF19 IFNGR2 SERPING1 IRF1 MT2A LCP2 TRIM25 C1S
g
Evidence and prior RA knowledge
Green: little prior RA evidence
Area: number of evidence layers
CD40
STAT4
SLAMF7
CD38
IL4I1
Integrated evidence score
Open Targets RA association score